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Q#1 A/An _____ topology is one in which all the edges are connected to another edge in a regular manner.

Options:

- A. Bounded
- B. Toroidal
- C. Non-Bounded
- D. Infinite Plane

Correct Option: B

Page Number(s): 244 (Module 131)

Ref Sentence by VU Stars: A toroidal topology is one in which all of the edges are connected to another edge in a regular manner.

Q#2 In Von Neumann Neighborhood how many neighbors are there for each cell of square lattice.

Options:

- A. Three
- B. Two
- C. Five
- D. Four

Correct Option: D

Page Number(s): 243 (Module 131)

Ref Sentence by VU Stars: A von Neumann neighborhood of radius 1 is a lattice where each cell has four neighbors: up, down, left, and right.

Q#3 In Moore Neighborhood how many neighbors are there for each cell of square lattice.

Options:

- A. Six
- B. Four
- C. Two
- D. Eight

Correct Option: D

Page Number(s): 243 (Module 131)

Ref Sentence by VU Stars: A Moore neighborhood of radius 1 is a lattice in which each cell has eight neighbors in the eight directions that touch either a side or a corner.

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Q#4 A network in which each node is randomly connected to other individual node is called as _____.

Options:

- A. Lattice Network
- B. Random Network
- C. Small-world Network
- D. Scale-free Network

Correct Option: B

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: In random networks, each individual is randomly connected to other individuals.

Q#5 Small-world networks are made up of dense clusters of highly interconnected nodes joined by _____.

Options:

- A. No links
- B. A few long-distance links
- C. Many short-distance links
- D. Random links

Correct Option: B

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: Small-world networks are made up of dense clusters of highly interconnected nodes joined by a few long distance links between them.

Q#6 The average of all the pairwise distances between nodes in a network is called _____.

Options:

- A. Clustering Coefficient
- B. Average Path Length
- C. Network Density
- D. Degree Centrality

Correct Option: B

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: The average path length is the average of all the pairwise distances between nodes in a network.

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Q#7 The clustering coefficient of a network is the average fraction of a node's immediate neighbors who are also _____.

Options:

- A. Far from the node
- B. Neighbors of the node's other neighbors
- C. Connected to random nodes
- D. Isolated from the network

Correct Option: B

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: The clustering coefficient of a network is the average fraction of a node's immediate neighbors who are also neighbors of the node's other neighbors.

Q#8 In a bounded topology, agents are _____.

Options:

- A. Allowed to move freely in any direction
- B. Not allowed to move beyond the edges of the world
- C. Allowed to wrap around the edges
- D. Allowed to expand the world

Correct Option: B

Page Number(s): 244 (Module 131)

Ref Sentence by VU Stars: A bounded topology is one in which agents are not allowed to move beyond the edges of the world.

Q#9 Which topology allows agents to reappear on the opposite side when they move off the world?

Options:

- A. Bounded Topology
- B. Infinite-plane Topology
- C. Toroidal Topology
- D. Network Topology

Correct Option: C

Page Number(s): 244 (Module 131)

Ref Sentence by VU Stars: A toroidal topology is one in which all of the edges are connected to another

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edge in a regular manner. In a rectangular lattice, the left side of the world is connected to the right side, while the top of the world is connected to the bottom.

Q#10 Which topology has no bounds and agents can keep moving forever in any direction?

Options:

- A. Bounded Topology
- B. Infinite-plane Topology
- C. Toroidal Topology
- D. Lattice Topology

Correct Option: B

Page Number(s): 244 (Module 131)

Ref Sentence by VU Stars: Finally, an infinite-plane topology is one where there are no bounds. In other words, agents can keep moving and moving in any direction forever.

Q#11 Which of the following is NOT a standard network topology used in ABM?

Options:

- A. Random Network
- B. Scale-free Network
- C. Small-world Network
- D. Toroidal Network

Correct Option: D

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: Besides the regular networks described earlier, the three most common network topologies are random, scale-free, and small-world.

Q#12 Scale-free networks have the property that _____.

Options:

- A. All nodes have the same number of connections
- B. Any subnetwork has the same properties as the global network
- C. Nodes are randomly connected
- D. Nodes form dense clusters

Correct Option: B

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Ref Sentence by VU Stars: Scale-free networks have the property that any subnetwork of the global network has the same properties as the global network.

Q#13 In scale-free networks, a common way to create them is by ____.

Options:

- A. Randomly connecting all nodes
- B. Preferential attachment
- C. Removing existing nodes
- D. Creating dense clusters

Correct Option: B

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: A common way to create this type of network is by adding new nodes and links to the system so that extant nodes with a large number of links are more likely to receive new connections. This technique is sometimes called preferential attachment.

Q#14 Which network topology is also sometimes called hub-and-spoke?

Options:

- A. Random Network
- B. Lattice Network
- C. Scale-free Network
- D. Small-world Network

Correct Option: C

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: This method for network creation tends to produce networks with central nodes that have many radiating links; because of the resemblance to a bicycle wheel, this network structure is sometimes also called hub-and-spoke.

Q#15 3D environments enable model developers to explore complex systems which are ____.

Options:

- A. Only two-dimensional
- B. Irreducibly bound up with a third dimension
- C. Only network-based
- D. Only lattice-based

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Correct Option: B

Page Number(s): 246 (Module 133)

Ref Sentence by VU Stars: 3D environments enable model developers to explore complex systems which are irreducibly bound up with a third dimension as well as sometimes increasing the apparent physical realism to their models.

Q#16 In 3D NetLogo, the orientation of an agent is described by heading, pitch, and ____.

Options:

- A. Yaw
- B. Roll
- C. Tilt
- D. Angle

Correct Option: B

Page Number(s): 246 (Module 133)

Ref Sentence by VU Stars: In 3D, the orientation of an agent can no longer be described by its heading alone, we must also use pitch and roll to describe the orientation of the agent.

Q#17 Geographic Information Systems (GIS) are environments that record large amounts of data related to ____.

Options:

- A. Network connections
- B. Physical locations in the world
- C. Agent properties
- D. Random numbers

Correct Option: B

Page Number(s): 246 (Module 133)

Ref Sentence by VU Stars: Geographic Information Systems (GIS) are environments that record large amounts of data that are related to physical locations in the world.

Q#18 GIS can serve as an ____ for an ABM to operate in.

Options:

- A. Agent type
- B. Interaction topology

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C. Data structure

D. Scheduling mechanism

Correct Option: B

Page Number(s): 246 (Module 133)

Ref Sentence by VU Stars: GIS can serve as an interaction topology. The 3D model, Flocking 3D.

Q#19 There are five basic classes of interactions in ABMs. Which of the following is NOT one of them?

Options:

A. Agent-self

B. Environment-self

C. Agent-Agent

D. Observer-Agent

Correct Option: D

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: There are five basic classes of interactions that exist in ABMs: agent-self, environment-self, agent-agent, environment-environment, and agent-environment.

Q#20 Birth events are a classic type of _____ interaction in ABMs.

Options:

A. Agent-agent

B. Agent-environment

C. Agent-self

D. Environment-self

Correct Option: C

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: Agent-Self Interactions ... One classic type of self-agent interaction is birth. Birth events are a typical event in ABMs. In these, one agent creates another agent.

Q#21 When grass regrows in a model, this is an example of _____ interaction.

Options:

A. Agent-self

B. Agent-environment

C. Environment-self

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D. Agent-agent

Correct Option: C

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: Environment-Self interactions are when areas of the environment alter or change themselves... The classic example of an environment-self interaction is when the grass regrows.

Q#22 When wolves consume sheep in a predator-prey model, this is an example of _____ interaction.

Options:

- A. Agent-self
- B. Agent-environment
- C. Environment-self
- D. Agent-agent

Correct Option: D

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: Agent-Agent Interactions ... We saw a canonical example of agent-agent interactions in the Wolf Sheep Predation model when the wolves consume the sheep.

Q#23 Communication between agents is an example of _____ interaction.

Options:

- A. Agent-self
- B. Agent-environment
- C. Environment-environment
- D. Agent-agent

Correct Option: D

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: A final example of agent-agent interaction is communication. Agents can share information about their own state as well as that of the world around them.

Q#24 Diffusion of pheromone in the Ants model is an example of _____ interaction.

Options:

- A. Agent-self
- B. Agent-environment
- C. Environment-environment

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D. Agent-agent

Correct Option: C

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: Environment-Environment Interactions: Interactions between different parts of the environment are probably the least commonly used type of interaction in agent-based models. However, there are some common uses of environment-environment interactions: one of these is diffusion.

Q#25 When an ant examines the environment to look for food, this is an example of _____ interaction.

Options:

- A. Agent-self
- B. Agent-environment
- C. Environment-self
- D. Agent-agent

Correct Option: B

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: Agent-Environment interactions occur when the agent manipulates or examines part of the world in which it exists... A common type of agent-environment interaction involves agents observing the environment.

Q#26 The observer in an ABM is a _____ agent.

Options:

- A. Low-level
- B. High-level
- C. Mobile
- D. Stationary

Correct Option: B

Page Number(s): 248 (Module 135)

Ref Sentence by VU Stars: The observer is a high-level agent that is responsible for ensuring that the model runs and proceeds according to the steps developed by the model author.

Q#27 The observer issues commands to _____.

Options:

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- A. Only the environment
- B. Agents and the environment
- C. Only the agents
- D. Other observers

Correct Option: B

Page Number(s): 248 (Module 135)

Ref Sentence by VU Stars: The observer issues commands to agents and the environment, telling them to manipulate their data or to take certain actions.

Q#28 Which of the following is NOT a standard way to interface with an ABM?

Options:

- A. Buttons
- B. Sliders
- C. Switches
- D. Databases

Correct Option: D

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: The most common control mechanism is a button... Sliders enable the model user to select a particular value from a range of numerical values. Switches enable the user to turn various elements of a model off or on.

Q#29 Sliders in NetLogo enable the user to select a value from ____.

Options:

- A. A drop-down menu
- B. A range of numerical values
- C. A list of options
- D. A text input field

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: Sliders enable the model user to select a particular value from a range of numerical values.

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Q#30 Switches in NetLogo are accessed as global variables that are _____.

Options:

- A. Numeric variables
- B. Boolean variables
- C. String variables
- D. List variables

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: Switches enable the user to turn various elements of a model off or on. In the Code tab, they are also accessed as global variables, but they are Boolean variables.

Q#31 Choosers in NetLogo enable the user to select from _____.

Options:

- A. A range of numerical values
- B. A predefined drop-down menu
- C. A text input field
- D. A slider

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: Choosers enable a model user to select a choice from a predefined drop-down menu that the modeler has created.

Q#32 Monitors in NetLogo display _____.

Options:

- A. The value of a global variable or calculation
- B. A 2D graph
- C. A text description
- D. A list of agents

Correct Option: A

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: As for output controls, monitors display the value of a global variable or calculation updated several times a second.

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Q#33 Plots in NetLogo provide ____.

Options:

- A. A text output
- B. A 2D graph
- C. A 3D visualization
- D. A monitor value

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: Plots provide traditional 2D graphs enabling the user to observe the change of an output variable over time.

Q#34 Which of the following is a guideline for designing visualization of an ABM?

Options:

- A. Complicate the visualization
- B. Simplify the visualization
- C. Hide the components
- D. Avoid emphasizing the main point

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: In general, there are three guidelines that should be kept in mind whenever designing the visualization of an ABM: simplify, explain, and emphasize.

Q#35 Which of the following is NOT a guideline for designing visualization of an ABM?

Options:

- A. Simplify
- B. Explain
- C. Emphasize
- D. Complicate

Correct Option: D

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: In general, there are three guidelines that should be kept in mind whenever designing the visualization of an ABM: simplify, explain, and emphasize.

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Q#36 Interactive running of a model means _____.

Options:

- A. Running the model with a script
- B. The user can interact as the model is running
- C. Running the model without any user input
- D. Running the model only once

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: This kind of spur-of-the-moment control is called interactive running, because the user can interact as the model is running.

Q#37 Batch running of a model involves _____.

Options:

- A. The user controlling the model directly
- B. Writing a script to run the model many times
- C. Running the model only once
- D. Running the model without any parameters

Correct Option: B

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: In contrast to interactive running, another type of user running of a model is called batch running. With batch running, instead of controlling the model directly, the user writes a script to run the model many times.

Q#38 The SETUP procedure in a NetLogo model is usually used for _____.

Options:

- A. The main loop of the model
- B. Initialization of the model
- C. Updating the user interface
- D. Running the model multiple times

Correct Option: B

Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: First of all, there is usually an initialization procedure that creates the agents,

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initializes the environment, and readies the user interface. In NetLogo this procedure is usually called SETUP.

Q#39 The GO procedure in a NetLogo model describes ____.

Options:

- A. What happens in one time unit of the model
- B. The initialization of the model
- C. The setup of the user interface
- D. The batch running of the model

Correct Option: A

Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: The other main part of the schedule is what is often called the main loop, or in NetLogo, the GO procedure. The GO procedure describes what happens in one time unit (or tick) of the model.

Q#40 If a model uses an asynchronous update schedule, this means that ____.

Options:

- A. All agents update simultaneously
- B. When agents change their state, that state is immediately seen by other agents
- C. Agents cannot change their state
- D. The model cannot run

Correct Option: B

Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: If a model uses an Asynchronous update schedule, this means that when agents change their state, that state is immediately seen by other agents.

Q#41 In a synchronous update schedule, ____.

Options:

- A. Changes made to an agent are immediately seen by other agents
- B. Changes made to an agent are not seen by other agents until the next clock tick
- C. Agents cannot change their state
- D. The model runs faster

Correct Option: B

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Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: In a Synchronous update schedule changes made to an agent are not seen by other agents until the next clock tick — that is, all agents update simultaneously.

Q#42 Sequential actions in ABM involve ____.

Options:

- A. All agents acting independently
- B. Only one agent acting at a time
- C. Agents acting in parallel
- D. Agents acting simultaneously

Correct Option: B

Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: Sequential actions involve only one agent acting at a time while Parallel actions are those in which all agents act independently.

Q#43 Parallel actions in ABM involve ____.

Options:

- A. Only one agent acting at a time
- B. All agents acting independently
- C. Agents acting sequentially
- D. Agents cannot act in parallel

Correct Option: B

Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: Sequential actions involve only one agent acting at a time while Parallel actions are those in which all agents act independently.

Q#44 Simulated concurrency uses one processor to ____.

Options:

- A. Act as a single agent
- B. Simulate many agents acting in parallel
- C. Run the model only once
- D. Initialize the model

Correct Option: B

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Page Number(s): 251 (Module 136)

Ref Sentence by VU Stars: Simulated concurrency uses one processor to simulate many agents acting in parallel.

Q#45 Descriptive statistics provide numerical measures that _____.

Options:

- A. Summarize a large data set
- B. Make the data more complex
- C. Hide the data
- D. Increase the data size

Correct Option: A

Page Number(s): 252 (Module 138)

Ref Sentence by VU Stars: The general methodology behind descriptive statistics is to provide numerical measures that summarize a large data set and describe the data set in such a way that it is not necessary to examine every single value.

Q#46 In the Spread of Disease model, as population density increases, the time to full infection _____.

Options:

- A. Dramatically increases
- B. Dramatically decreases
- C. Remains the same
- D. Becomes unpredictable

Correct Option: B

Page Number(s): 252 (Module 139)

Ref Sentence by VU Stars: We conclude that as the population density increases, the time to full infection dramatically decreases.

Q#47 BehaviorSpace in NetLogo is a tool for _____.

Options:

- A. Creating agents
- B. Running the model multiple times with different settings
- C. Visualizing the model
- D. Debugging the model

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Correct Option: B

Page Number(s): 253 (Module 141)

Ref Sentence by VU Stars: In NetLogo there is a tool called BehaviorSpace. ABM toolkits are often full featured programming languages, allowing you to write your own tools for creating experiments to produce the data sets you want to analyze. These tools will automatically run the model multiple times with multiple different settings.

Q#48 A chaotic system is a deterministic system that is _____.

Options:

- A. Easy to predict
- B. Difficult to predict
- C. Always stable
- D. Never changes

Correct Option: B

Page Number(s): 288 (Module 164)

Ref Sentence by VU Stars: A chaotic system is a deterministic system that is difficult to predict.

Q#49 The term "chaos" in science has a precise mathematical meaning and refers to _____.

Options:

- A. Disorder or confusion
- B. A deterministic system that is difficult to predict
- C. A random system
- D. A stable system

Correct Option: B

Page Number(s): 288 (Module 164)

Ref Sentence by VU Stars: In science, chaos is an important conceptual paradox that has a precise mathematical meaning: A chaotic system is a deterministic system that is difficult to predict.

Q#50 A deterministic system is defined as one whose state at one time _____.

Options:

- A. Completely determines its state for all future times
- B. Has no effect on future states
- C. Is random

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D. Is unpredictable

Correct Option: A

Page Number(s): 288 (Module 164)

Ref Sentence by VU Stars: A deterministic system is defined as one whose state at one time completely determines its state for all future times.

Q#51 The butterfly effect is associated with _____.

Options:

A. Linear systems

B. Chaotic systems

C. Stable systems

D. Random systems

Correct Option: B

Page Number(s): 288 (Module 164)

Ref Sentence by VU Stars: Butterfly effect (associated with chaotic systems).

Q#52 A complex adaptive system (CAS) is a system that _____.

Options:

A. Never changes its behavior

B. Changes its behavior in response to its environment

C. Is always stable

D. Has no interactions

Correct Option: B

Page Number(s): 287 (Module 162)

Ref Sentence by VU Stars: An adaptive system (or a complex adaptive system, CAS) is a system that changes its behavior in response to its environment.

Q#53 The adaptive change that occurs in a CAS is often relevant to _____.

Options:

A. Achieving a goal or objective

B. Creating chaos

C. Reducing complexity

D. Eliminating agents

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Correct Option: A

Page Number(s): 287 (Module 162)

Ref Sentence by VU Stars: The adaptive change that occurs is often relevant to achieving a goal or objective.

Q#54 The systems approach stands on the assumption that breaking down a complex concept into simple units helps in ____.

Options:

- A. Making it more complex
- B. Better understanding of the complexity
- C. Eliminating the system
- D. Creating chaos

Correct Option: B

Page Number(s): 287 (Module 162)

Ref Sentence by VU Stars: The approach stands on the assumption that breaking down of a complex concept into simple easy to understand units helps in better understanding of the complexity.

Q#55 Ludwig von Bertalanffy first proposed the 'General System Theory' as ____.

Options:

- A. A theory of chaos
- B. A theory of systems
- C. A theory of randomness
- D. A theory of agents

Correct Option: B

Page Number(s): 287 (Module 162)

Ref Sentence by VU Stars: Ludwig von Bertalanffy first proposed it as 'General System Theory'.

Q#56 The CABC framework has how many levels for Agent-based modeling?

Options:

- A. Two
- B. Three
- C. Four
- D. Five

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Correct Option: B

Page Number(s): 289 (Module 168)

Ref Sentence by VU Stars: CABC framework has 3 levels for Agent-based modeling.

Q#57 Which of the following is a level of the CABC framework?

Options:

- A. Exploratory Agent-based modeling (EABM)
- B. Descriptive Agent-based Modeling (DREAM)
- C. Validated Agent-based Modeling (VABM)
- D. All of the above

Correct Option: D

Page Number(s): 289 (Module 168)

Ref Sentence by VU Stars: CABC framework has 3 levels for Agent-based modeling: Exploratory Agent-based modeling (EABM), Descriptive Agent-based Modeling (DREAM), Validated Agent-based Modeling (VABM).

Q#58 DREAM stands for _____.

Options:

- A. Descriptive Agent-based Modeling
- B. Dynamic Agent-based Modeling
- C. Distributed Agent-based Modeling
- D. Deterministic Agent-based Modeling

Correct Option: A

Page Number(s): 289 (Module 168)

Ref Sentence by VU Stars: Descriptive Agent-based Modeling (DREAM).

Q#59 VOMAS stands for _____.

Options:

- A. Virtual Overlay Multiagent System
- B. Validated Overlay Multiagent System
- C. Virtual Operating Multiagent System
- D. Validated Operating Multiagent System

Correct Option: A

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Page Number(s): 289 (Module 168)

Ref Sentence by VU Stars: Virtual Overlay Multiagent System (VOMAS).

Q#60 A graph and a network are ____.

Options:

- A. Completely different entities
- B. The same entity with different terms in different domains
- C. Always directed
- D. Always undirected

Correct Option: B

Page Number(s): 289 (Module 169)

Ref Sentence by VU Stars: Graph/Network: Same entity different terms in different domains.

Q#61 A network can also be referred to as a ____.

Options:

- A. Weighted graph
- B. Unweighted graph
- C. Directed graph
- D. Undirected graph

Correct Option: A

Page Number(s): 289 (Module 169)

Ref Sentence by VU Stars: Network = weighted graph.

Q#62 Which of the following is a benefit of complex network-based modeling?

Options:

- A. Very well-defined
- B. Lot of free tools
- C. Well-defined mathematical models
- D. All of the above

Correct Option: D

Page Number(s): 289 (Module 169)

Ref Sentence by VU Stars: Key benefits: Very well-defined, Lot of free tools, Well-defined mathematical models.

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Q#63 Which of the following is a limitation of complex network-based modeling?

Options:

- A. Well-defined mathematical models
- B. Difficulty in graph mining
- C. Lot of free tools
- D. Very well-defined

Correct Option: B

Page Number(s): 289 (Module 169)

Ref Sentence by VU Stars: Cons: Difficulty in graph mining – lack of expertise in the community.

Q#64 In Bernoulli distribution, each trial has ____.

Options:

- A. Multiple possible outcomes
- B. One out of two possible outcomes (success or failure)
- C. An infinite number of outcomes
- D. No outcomes

Correct Option: B

Page Number(s): 290 (Module 172)

Ref Sentence by VU Stars: Bernoulli trail is one of the simplest experiment you can do in statistic and probability which has one out of two possible outcome.

Q#65 The expected value of a Bernoulli random variable X is ____.

Options:

- A. p
- B. 1-p
- C. p(1-p)
- D. 0

Correct Option: A

Page Number(s): 290 (Module 172)

Ref Sentence by VU Stars: Below is the expected value of random variable X is calculated: $E(X) = p$.

Q#66 The variance of a Bernoulli random variable X is ____.

Options:

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- A. p
- B. 1-p
- C. $p(1-p)$
- D. 0

Correct Option: C

Page Number(s): 290 (Module 172)

Ref Sentence by VU Stars: Below is the variance of the random variable X is calculated as: $V(X) = p(1-p)$.

Q#67 Binomial distribution is the discrete probability distribution with parameters n and p of the number of successes in _____.

Options:

- A. A single trial
- B. A sequence of n independent experiments
- C. An infinite sequence
- D. A random sequence

Correct Option: B

Page Number(s): 290 (Module 173)

Ref Sentence by VU Stars: Binomial distribution is the discrete probability distribution with parameters n and p of the number of successes in the sequence of n independent experiments.

Q#68 The mean of a binomial distribution is _____.

Options:

- A. np
- B. $np(1-p)$
- C. p
- D. 1-p

Correct Option: A

Page Number(s): 290 (Module 173)

Ref Sentence by VU Stars: Expected value $E(X) = np$.

Q#69 The variance of a binomial distribution is _____.

Options:

- A. np

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B. $np(1-p)$

C. p

D. $1-p$

Correct Option: B

Page Number(s): 290 (Module 173)

Ref Sentence by VU Stars: Variance $V(X) = np(1-p)$.

Q#70 Poisson distribution gives the probability of independent events occurring in _____.

Options:

A. A fixed interval of time

B. An infinite interval of time

C. A random interval of time

D. No interval of time

Correct Option: A

Page Number(s): 290 (Module 174)

Ref Sentence by VU Stars: A discrete frequency distribution that gives the probability of independent events occurring in a fixed interval of time is known as the Poisson distribution.

Q#71 An important property of Poisson distribution is that _____.

Options:

A. Mean is greater than variance

B. Mean is less than variance

C. Mean and variance are equal

D. Mean and variance are not related

Correct Option: C

Page Number(s): 290 (Module 174)

Ref Sentence by VU Stars: This is the important property of Poisson distribution that is mean and variance is equal to λ .

Q#72 The uniform distribution is also known as _____.

Options:

A. Rectangular distribution

B. Triangular distribution

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C. Exponential distribution

D. Normal distribution

Correct Option: A

Page Number(s): 291 (Module 175)

Ref Sentence by VU Stars: Here, We will understand "Uniform Distribution" also known as "rectangular distribution".

Q#73 The probability density function for a uniform distribution always takes a random number on _____.

Options:

A. $[0, 1]$

B. $[0, \infty]$

C. $[-\infty, \infty]$

D. $[1, 2]$

Correct Option: A

Page Number(s): 291 (Module 175)

Ref Sentence by VU Stars: The uniform distribution is a continuous distribution. It takes values within a specified range. Probability density function for a uniform distribution always take a random number on $[0,1]$.

Q#74 The Exponential Distribution is a _____ valued probability distribution.

Options:

A. Discrete

B. Continuous

C. Binary

D. Integer

Correct Option: B

Page Number(s): 291 (Module 176)

Ref Sentence by VU Stars: The Exponential Distribution is a continuous valued probability distribution.

Q#75 In Exponential Distribution, λ is called the _____.

Options:

A. Rate parameter

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- B. Shape parameter
- C. Scale parameter
- D. Location parameter

Correct Option: A

Page Number(s): 291 (Module 176)

Ref Sentence by VU Stars: Here λ is called the rate parameter.

Q#76 The inverse transform technique is used when _____.

Options:

- A. The cdf is very complex
- B. The form of cdf is so simple that its f^{-1} can be computed easily
- C. The distribution is discrete
- D. The distribution is unknown

Correct Option: B

Page Number(s): 291 (Module 176)

Ref Sentence by VU Stars: The inverse transform technique is used when the form of cdf is so simple that its f^{-1} can be computed easily.

Q#77 A triangular distribution is a continuous probability distribution with a probability density function shaped like a _____.

Options:

- A. Rectangle
- B. Triangle
- C. Circle
- D. Bell

Correct Option: B

Page Number(s): 292 (Module 177)

Ref Sentence by VU Stars: A triangular distribution is a continuous probability distribution with a probability density function shaped like a triangle.

Q#78 A triangular distribution is defined by _____.

Options:

- A. Minimum value a, maximum value b, peak value c

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- B. Mean and variance
- C. Rate parameter only
- D. Shape parameter only

Correct Option: A

Page Number(s): 292 (Module 177)

Ref Sentence by VU Stars: A triangular distribution is a continuous probability distribution with a probability density function shaped like a triangle defined by: The minimum value a, The maximum value b, The peak value c.

Q#79 A Customer in a queueing system is _____.

Options:

- A. Anything that visits a facility and requires some service
- B. Only a person
- C. Only a machine
- D. Only a computer

Correct Option: A

Page Number(s): 292 (Module 178)

Ref Sentence by VU Stars: A Customer is anything that visits a facility and requires some service.

Q#80 The calling population is the _____.

Options:

- A. Population of potential customers
- B. Population of servers
- C. Population of machines
- D. Population of computers

Correct Option: A

Page Number(s): 293 (Module 179)

Ref Sentence by VU Stars: Calling population is the population of potential customers.

Q#81 A calling population can be _____.

Options:

- A. Finite or infinite
- B. Only finite

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- C. Only infinite
- D. Neither finite nor infinite

Correct Option: A

Page Number(s): 293 (Module 179)

Ref Sentence by VU Stars: Calling population is the population of potential customers. Finite or infinite.

Q#82 The arrival rate is the main difference between ____.

Options:

- A. Finite and infinite population models
- B. Single server and multiple server models
- C. FIFO and LIFO models
- D. M/M/1 and M/M/c models

Correct Option: A

Page Number(s): 293 (Module 179)

Ref Sentence by VU Stars: Arrival rate is the main difference between the finite and infinite population models.

Q#83 Every queueing system has some ____.

Options:

- A. Capacity
- B. Server
- C. Customer
- D. Queue

Correct Option: A

Page Number(s): 293 (Module 180)

Ref Sentence by VU Stars: Every queueing system has some capacity.

Q#84 If the entering system finds the capacity full, it will ____.

Options:

- A. Wait indefinitely
- B. Return to the calling population
- C. Create more capacity
- D. Remove other customers

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Correct Option: B

Page Number(s): 293 (Module 180)

Ref Sentence by VU Stars: If the entering system finds the capacity full it will return to the calling population.

Q#85 Poisson arrival process is the most important for _____.

Options:

- A. Scheduled arrivals
- B. Random arrivals
- C. Constant arrivals
- D. Infinite arrivals

Correct Option: B

Page Number(s): 293 (Module 181)

Ref Sentence by VU Stars: Poisson arrival process is the most important for random arrivals.

Q#86 In Poisson arrival process, the number of arrivals in a time interval of length t has _____.

Options:

- A. Exponential distribution with mean λt
- B. Poisson distribution with mean λt
- C. Normal distribution with mean λt
- D. Uniform distribution with mean λt

Correct Option: B

Page Number(s): 293 (Module 181)

Ref Sentence by VU Stars: So the number of arrivals in a time interval of length t has Poisson distribution with mean λt customers.

Q#87 Queue behavior refers to _____.

Options:

- A. The actions of the customer while waiting in the queue
- B. The logical ordering of customers in the queue
- C. The service time distribution
- D. The number of servers

Correct Option: A

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Page Number(s): 294 (Module 182)

Ref Sentence by VU Stars: Queue Behavior: The actions of the customer while waiting in the queue system for his service is known as the queue behaviour.

Q#88 Queue discipline refers to _____.

Options:

- A. The actions of the customer while waiting in the queue
- B. The logical ordering of the customers in a queue
- C. The service time distribution
- D. The number of servers

Correct Option: B

Page Number(s): 294 (Module 182)

Ref Sentence by VU Stars: Queue Discipline: The logical ordering of the customers in a queue and determining that which customer should come for service when the server is free.

Q#89 Which of the following is a common queue discipline?

Options:

- A. FIFO
- B. LIFO
- C. SIRO
- D. All of the above

Correct Option: D

Page Number(s): 294 (Module 182)

Ref Sentence by VU Stars: FIFO, LIFO, SIRO, SPT and PR are some common queue disciplines.

Q#90 In Kendall's queueing notation, "A" represents _____.

Options:

- A. The service-time distribution
- B. The inter-arrival-time distribution
- C. The number of parallel servers
- D. The system capacity

Correct Option: B

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Page Number(s): 294 (Module 184)

Ref Sentence by VU Stars: "A" represents the inter-arrival-time distribution.

Q#91 In Kendall's queueing notation, "B" represents ____.

Options:

- A. The inter-arrival-time distribution
- B. The service-time distribution
- C. The number of parallel servers
- D. The system capacity

Correct Option: B

Page Number(s): 294 (Module 184)

Ref Sentence by VU Stars: "B" represents the service-time distribution.

Q#92 In Kendall's queueing notation, "M" represents ____.

Options:

- A. Exponential or Markov
- B. Constant or deterministic
- C. Arbitrary or general
- D. Erlang

Correct Option: A

Page Number(s): 294 (Module 184)

Ref Sentence by VU Stars: Common symbols for A and B include M (exponential or Markov).

Q#93 In Kendall's queueing notation, "D" represents ____.

Options:

- A. Exponential or Markov
- B. Constant or deterministic
- C. Arbitrary or general
- D. Erlang

Correct Option: B

Page Number(s): 294 (Module 184)

Ref Sentence by VU Stars: D (constant or deterministic).

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Q#94 The two important statistical properties of random numbers are _____.

Options:

- A. Uniformity and independence
- B. Uniformity and dependence
- C. Randomness and independence
- D. Randomness and dependence

Correct Option: A

Page Number(s): 295 (Module 186)

Ref Sentence by VU Stars: Uniformity and independence are two important statistical properties.

Q#95 A random number should be drawn from _____.

Options:

- A. Continuous uniform distribution between 0 and 1
- B. Discrete uniform distribution between 0 and 1
- C. Normal distribution between 0 and 1
- D. Exponential distribution between 0 and 1

Correct Option: A

Page Number(s): 295 (Module 186)

Ref Sentence by VU Stars: Each sample must be drawn from continuous uniform distribution between 0 and 1.

Q#96 The expected value of a random number R_i is _____.

Options:

- A. 0
- B. 1
- C. $1/2$
- D. $1/3$

Correct Option: C

Page Number(s): 295 (Module 186)

Ref Sentence by VU Stars: The density function and the expected value of R_i is: $E(R_i) = 1/2$.

Q#97 The variance of a random number R_i is _____.

Options:

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A. 1/2

B. 1/4

C. 1/12

D. 1/3

Correct Option: C

Page Number(s): 295 (Module 186)

Ref Sentence by VU Stars: And the variance is: $V(R_i) = 1/12$.

Q#98 The Linear Congruential Method is a widely used technique that _____.

Options:

A. Yields sequences with shorter periods

B. Yields sequences with longer periods

C. Produces truly random numbers

D. Cannot be replicated

Correct Option: B

Page Number(s): 295 (Module 187)

Ref Sentence by VU Stars: Linear Congruential Method: Widely used technique. Yields sequences with longer periods.

Q#99 In the Linear Congruential Method, when $c = 0$, it is known as _____.

Options:

A. Mixed congruential method

B. Multiplicative congruential method

C. Additive congruential method

D. Exponential congruential method

Correct Option: B

Page Number(s): 295 (Module 187)

Ref Sentence by VU Stars: This is called mixed congruential method. When $c = 0$ then known as multiplicative congruential.

Q#100 In the Linear Congruential Method, when $c \neq 0$, it is known as _____.

Options:

A. Mixed congruential method

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- B. Multiplicative congruential method
- C. Additive congruential method
- D. Exponential congruential method

Correct Option: A

Page Number(s): 295 (Module 187)

Ref Sentence by VU Stars: This is called mixed congruential method. When $c = 0$ then known as multiplicative congruential.

Q#101 Pseudo-random numbers are numbers generated that _____.

Options:

- A. Are truly random
- B. Simulate the properties of uniform distribution and independence
- C. Cannot be replicated
- D. Have no pattern

Correct Option: B

Page Number(s): 296 (Module 188)

Ref Sentence by VU Stars: Producing numbers between 0 and 1 that simulates the properties of uniform distribution and independence.

Q#102 Tests for random numbers are divided into two categories: _____.

Options:

- A. Uniformity and independence
- B. Uniformity and dependence
- C. Randomness and independence
- D. Randomness and dependence

Correct Option: A

Page Number(s): 296 (Module 189)

Ref Sentence by VU Stars: Tests for Random numbers: Two categories: 1. Uniformity. 2. Independence.

Q#103 A level of significance α is the probability of _____.

Options:

- A. Accepting the null hypothesis
- B. Rejecting the null hypothesis

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C. Accepting the alternative hypothesis

D. Rejecting the alternative hypothesis

Correct Option: B

Page Number(s): 296 (Module 189)

Ref Sentence by VU Stars: A level of significance α must be stated for each test. Where α is probability of rejecting the null hypothesis.

Q#104 A random variate is a generated variable using _____.

Options:

A. Truly random numbers

B. Uniformly distributed pseudorandom numbers

C. Exponentially distributed pseudorandom numbers

D. Normally distributed pseudorandom numbers

Correct Option: B

Page Number(s): 296 (Module 190)

Ref Sentence by VU Stars: A random variate is a generated variable using uniformly distributed pseudorandom numbers.

Q#105 Which of the following is a technique for generating random variates?

Options:

A. Inverse-transform technique

B. Acceptance-rejection technique

C. Both A and B

D. Neither A nor B

Correct Option: C

Page Number(s): 296 (Module 190)

Ref Sentence by VU Stars: This topic includes: inverse-transform technique, Acceptance-rejection technique.

Q#106 The inverse-transform technique can be used to sample from which distribution?

Options:

A. Exponential

B. Uniform

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- C. Weibull
- D. All of the above

Correct Option: D

Page Number(s): 296 (Module 191)

Ref Sentence by VU Stars: The inverse-transform technique can be used to sample from the exponential, The uniform, The Weibull, The triangular and empirical distributions.

Q#107 The basic principle of inverse-transform technique is to find _____.

Options:

- A. The inverse function of F, F^{-1}
- B. The derivative of F
- C. The integral of F
- D. The limit of F

Correct Option: A

Page Number(s): 296 (Module 191)

Ref Sentence by VU Stars: The basic principle is to find the inverse function of F, F^{-1} such that $FF^{-1}=I$.

Q#108 The acceptance-rejection technique is used to _____.

Options:

- A. Generate random variates
- B. Accept all random numbers
- C. Reject all random numbers
- D. Generate only uniform random numbers

Correct Option: A

Page Number(s): 297 (Module 192)

Ref Sentence by VU Stars: Acceptance Rejection Technique is a method used to generate random variate.

Q#109 The efficiency of an acceptance-rejection technique rests on _____.

Options:

- A. Increasing the total number of rejections
- B. Reducing the total number of rejections
- C. Accepting all random numbers
- D. Rejecting all random numbers

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Correct Option: B

Page Number(s): 297 (Module 192)

Ref Sentence by VU Stars: The efficiency of an acceptance-rejection technique rest on reducing the total of rejections.

Q#110 Direct transformation is used for generating which distribution?

Options:

- A. Uniform distribution
- B. Normal and lognormal distributions
- C. Exponential distribution
- D. Triangular distribution

Correct Option: B

Page Number(s): 297 (Module 193)

Ref Sentence by VU Stars: Direct transformation is used for the normal and lognormal distributions.

Q#111 The Box and Muller method is used for generating ____.

Options:

- A. Uniform random numbers
- B. Normal random variates
- C. Exponential random variates
- D. Triangular random variates

Correct Option: B

Page Number(s): 297 (Module 193)

Ref Sentence by VU Stars: Box and Muller made this method and is easy to program in scientific programming languages.

Q#112 The probability distribution of a sum of two or more independent random variables is called ____.

Options:

- A. Direct transformation
- B. Convolution method
- C. Acceptance-rejection method
- D. Inverse-transform method

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Correct Option: B

Page Number(s): 297 (Module 193)

Ref Sentence by VU Stars: The probability distribution of a sum of two or more independent random variables is called convolution method.

Q#113 Histograms are also known as _____.

Options:

- A. Frequency distribution
- B. Probability distribution
- C. Cumulative distribution
- D. Density function

Correct Option: A

Page Number(s): 298 (Module 196)

Ref Sentence by VU Stars: Histograms is a powerful technique for identifying how a distribution is shaped. Histogram is also known as frequency distribution.

Q#114 The selection of a family of distributions is based on _____.

Options:

- A. The context of what is being investigated
- B. The shape of the histogram
- C. Both A and B
- D. Neither A nor B

Correct Option: C

Page Number(s): 298 (Module 197)

Ref Sentence by VU Stars: The selection of a family of distributions is based on the context for what it is being investigated. The shape of the histogram.

Q#115 The Quantile-Quantile plot is a graphical method for _____.

Options:

- A. Estimating parameters
- B. Comparing two probability distributions by plotting their quantiles against each other
- C. Creating histograms
- D. Calculating the mean

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Correct Option: B

Page Number(s): 298 (Module 198)

Ref Sentence by VU Stars: A Quantile-Quantile is a graphical method for comparison of two probability distributions by plotting their qualities against each other.

Q#116 After selecting the family of distribution, the next step is ____.

Options:

- A. Data collection
- B. Parameter estimation
- C. Model verification
- D. Model validation

Correct Option: B

Page Number(s): 299 (Module 199)

Ref Sentence by VU Stars: After selecting the family of distribution, the next step is to estimate the parameters of the distribution.

Q#117 The sample mean is denoted by ____.

Options:

- A. $\bar{x}$
- B. s^2
- C. σ^2
- D. μ

Correct Option: A

Page Number(s): 299 (Module 200)

Ref Sentence by VU Stars: $\bar{x}$ = sample mean.

Q#118 The sample variance is denoted by ____.

Options:

- A. $\bar{x}$
- B. s^2
- C. σ^2
- D. μ

Correct Option: B

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Page Number(s): 299 (Module 200)

Ref Sentence by VU Stars: The sample variance s^2 is defined as.

Q#119 ENIAC was the first computer system with _____.

Options:

- A. A good operating system
- B. No operating system
- C. A database system
- D. A network system

Correct Option: B

Page Number(s): 300 (Module 201)

Ref Sentence by VU Stars: ENIAC was the first computer system with no operating system and was large in size.

Q#120 RISC stands for _____.

Options:

- A. Reduced Instruction Set Computer
- B. Random Instruction Set Computer
- C. Regular Instruction Set Computer
- D. Reliable Instruction Set Computer

Correct Option: A

Page Number(s): 300 (Module 201)

Ref Sentence by VU Stars: Reduced instruction set computer (RISC).

Q#121 CISC stands for _____.

Options:

- A. Complex Instruction Set Computer
- B. Compound Instruction Set Computer
- C. Central Instruction Set Computer
- D. Comprehensive Instruction Set Computer

Correct Option: A

Page Number(s): 300 (Module 201)

Ref Sentence by VU Stars: Complex instruction set computer (CISC).

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Q#122 ACID properties in database systems include _____.

Options:

- A. Atomicity, Consistency, Isolation, Durability
- B. Atomicity, Concurrency, Isolation, Durability
- C. Availability, Consistency, Isolation, Durability
- D. Atomicity, Consistency, Integrity, Durability

Correct Option: A

Page Number(s): 301 (Module 202)

Ref Sentence by VU Stars: "ACID" properties.

Q#123 An operating system is the most privileged of _____ on the system.

Options:

- A. Hardware elements
- B. Software elements
- C. Firmware elements
- D. Database elements

Correct Option: B

Page Number(s): 301 (Module 203)

Ref Sentence by VU Stars: The most privileged of software elements on the system.

Q#124 ARPANET was an early _____.

Options:

- A. Operating system
- B. Database system
- C. Computer network
- D. Programming language

Correct Option: C

Page Number(s): 302 (Module 204)

Ref Sentence by VU Stars: ARPANET was an early computer network.

Q#125 LAN stands for _____.

Options:

- A. Local Area Network

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- B. Large Area Network
- C. Logical Area Network
- D. Linear Area Network

Correct Option: A

Page Number(s): 302 (Module 204)

Ref Sentence by VU Stars: Local area network was introduced in 1970s.

Q#126 Wireless technology for networks began in the 1970s with _____ as foundation.

Options:

- A. Ethernet
- B. ARPANET
- C. Aloha net
- D. Token ring

Correct Option: C

Page Number(s): 302 (Module 204)

Ref Sentence by VU Stars: Wireless technology has improved. Began in 1970s with Aloha net as foundation.

Q#127 Performance evaluation is needed when _____.

Options:

- A. Selecting a specific computer architecture
- B. Selecting an operating system
- C. Selecting a database system
- D. All of the above

Correct Option: D

Page Number(s): 302 (Module 205)

Ref Sentence by VU Stars: Selecting a specific computer architecture for an application. An operating system. A database system. A wide area or local area network system.

Q#128 A model provides a tool to _____.

Options:

- A. Define a system
- B. Study a system before its existence

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- C. Both A and B
- D. Neither A nor B

Correct Option: C

Page Number(s): 303 (Module 207)

Ref Sentence by VU Stars: Models provide a tool to define a system. This provide an environment to study a system before its existence.

Q#129 Modeling is better if _____.

Options:

- A. Physical laws are available
- B. Pictorial representation is possible
- C. Input/output elements are manageable
- D. All of the above

Correct Option: D

Page Number(s): 303 (Module 207)

Ref Sentence by VU Stars: Modeling is better if: Physical laws are available. Pictorial representation is possible. Input/output elements are manageable.

Q#130 The goal of comparing two system designs is to obtain _____.

Options:

- A. Point estimates
- B. Interval estimates of the difference in mean performance
- C. Both A and B
- D. Neither A nor B

Correct Option: C

Page Number(s): 304 (Module 209)

Ref Sentence by VU Stars: The goal of the simulation experiments is to obtain point and interval estimates of the difference in mean performance.

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Q#1 _____ enables the model user to select a value from a range of numerical values.

Options:

- A. Sliders
- B. Buttons
- C. Input Boxes
- D. Switches

Correct Option: A

Page Number(s): 250 (Module 135)

Ref Sentence by VU Stars: Sliders enable the model user to select a particular value from a range of numerical values.

Q#2 _____ networks are made up of dense clusters of highly interconnected nodes joined by a few long-distance links between them.

Options:

- A. Random Network
- B. Lattice Network
- C. Small-world Network
- D. Scale-free Network

Correct Option: C

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: Small-world networks are made up of dense clusters of highly interconnected nodes joined by a few long distance links between them.

Q#3 Communication is an example of _____ interaction.

Options:

- A. Agent – self
- B. Agent – Environment
- C. Environment – Environment
- D. Agent – Agent

Correct Option: D

Page Number(s): 247 (Module 134)

Ref Sentence by VU Stars: A final example of agent-agent interaction is communication.

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Q#4 Agents that cannot move at the landscape are known as _____.

Options:

- A. Mobile Agents
- B. Model-based Agents
- C. Stationary Agents
- D. Connecting Agents

Correct Option: C

Page Number(s): 238 (Module 126)

Ref Sentence by VU Stars: stationary agents that cannot move at all; and connecting agents that link two or more other agents.

Q#5 Model _____ is the implementation by one researchers of a conceptual model previously implemented by someone else.

Options:

- A. Verification
- B. Validation
- C. Replication
- D. Correctness

Correct Option: C

Page Number(s): 271 (Module 145)

Ref Sentence by VU Stars: Model replication is the implementation by one researcher or group of researchers of a conceptual model previously implemented by someone else.

Q#6 A/An _____ topology is one in which all the edges are connected to another edge in a regular manner.

Options:

- A. Bounded
- B. Toroidal
- C. Non-Bounded
- D. Infinite Plane

Correct Option: B

Page Number(s): 244 (Module 131)

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Ref Sentence by VU Stars: A toroidal topology is one in which all of the edges are connected to another edge in a regular manner.

Q#7 In Von Neumann Neighborhood how many neighbors are there for each cell of square lattice.

Options:

- A. Three
- B. Two
- C. Five
- D. Four

Correct Option: D

Page Number(s): 243 (Module 131)

Ref Sentence by VU Stars: A von Neumann neighborhood ... of radius 1 is a lattice where each cell has four neighbors: up, down, left, and right.

Q#8 A network in which each node is randomly connected to other individual node is called as ____.

Options:

- A. Lattice Network
- B. Random Network
- C. Small-world Network
- D. Scale-free Network

Correct Option: B

Page Number(s): 245 (Module 132)

Ref Sentence by VU Stars: In random networks, each individual is randomly connected to other individuals.

Q#9 For different Models the granularity is ____.

Options:

- A. Multiple
- B. Equal
- C. Different
- D. Similar

Correct Option: C

Page Number(s): 242 (Module 127)

Ref Sentence by VU Stars: So how do you choose the level of complexity for the agents in your model? ...

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These two models highlight the relationship between the question being investigated, the interactions being modeled, and the resulting choices of granularity for the agents.

Q#10 Which Statement is not true for observers

Options:

- A. It is a low-level agent
- B. The observer issues commands to agents and the environment
- C. It ensures that the model runs and proceeds according to the steps developed by the model author
- D. It is a high-level agent

Correct Option: A

Page Number(s): 248 (Module 135)

Ref Sentence by VU Stars: The observer is a high-level agent that is responsible for ensuring that the model runs and proceeds according to the steps developed by the model author. The observer issues commands to agents and the environment, telling them to manipulate their data or to take certain actions.

Q#11 There are ____ keyways of doing model documentation.

Options:

- A. Two
- B. Three
- C. Four
- D. Five

Correct Option: A

Page Number(s): 278 (Module 159)

Ref Sentence by VU Stars: (This specific answer may not be explicitly stated as 'two keyways' in the provided handouts, but the context of documentation and reporting typically refers to two types: program and progress documentation.) **Based on the handout context, there are two types of documentation: Program and Progress.** (Found in Module 20: "There are two types of documentation: Program and Progress.")

Q#12 Agent-based methods may be particularly valuable in thesciences.

Options:

- A. SOCIAL
- B. NONE
- C. BIOLOGICAL

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D. PHYSICAL

Correct Option: A

Page Number(s): 219 (Module 80)

Ref Sentence by VU Stars: Agent-based methods may be particularly valuable in the social sciences.

Q#13 In Moore Neighborhood how many neighbors are there for each cell of square lattice.

Options:

- A. Six
- B. Four
- C. Two
- D. Eight

Correct Option: D

Page Number(s): 243 (Module 131)

Ref Sentence by VU Stars: A Moore neighborhood ... of radius 1 is a lattice in which each cell has eight neighbors in the eight directions that touch either a side or a corner: up, down, left, right, up-left, up-right, down-left, and downright.

Q#14 Agent cognition is defined as how much _____ is added into the agents.

Options:

- A. Complexity
- B. Granularity
- C. Coherence
- D. Simplicity

Correct Option: A

Page Number(s): 241 (Module 128)

Ref Sentence by VU Stars: As we have described, agents have different properties and behaviors. Still, how do agents examine their properties and the world around them to decide what actions to take? This question is resolved by a decision-making process called agent cognition.

Q#15 In agent-based modeling the environment consists of _____ surrounding the agents as they act and interact within the model.

Options:

- A. Habitats and decisions

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B. Conditions

C. Conditions & habitats

D. Conditions and decisions

Correct Option: C

Page Number(s): 242 (Module 130)

Ref Sentence by VU Stars: The environment consists of the conditions and habitats surrounding the agents as they act and interact within the model.



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